The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 December 1997

INCH POUND

MIL-PRF-19500/253E
27 September 1997
SUPERSEDING
MIL-S-19500/253D
10 September 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPE 2N930 JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, low-power transistors. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See 3.3 (TO-18).

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}
mW	mW	V dc	V dc	V dc	mA dc	°C
300	600	60	45	6	30	-55 to +200

1/ Derate linearly at 2.0 mW/°C above T_A = +25°C.
2/ Derate linearly at 4.0 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE1} 1/	h_{FE2} 1/	C_{obo}	$\|h_{fe}\|$	$V_{BE(SAT)}$ 1/	$V_{CE(SAT)}$ 1/
	V_{CE} = 5 V dc I_C = 10 µA dc	V_{CE} = 5 V dc I_C = 500 µA dc	V_{CB} = 5 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 5 V dc I_C = 500 µA dc f = 30 MHz	I_C = 10 mA dc I_B = 0.5 mA dc	I_C = 10 mA dc I_B = 0.5 mA dc
			pF		V dc	V dc
Min	100	150		1.5	0.6	
Max	300		8.0	6.0	1.0	1.0

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated performance specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500 and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Devices shall be marked in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.43.

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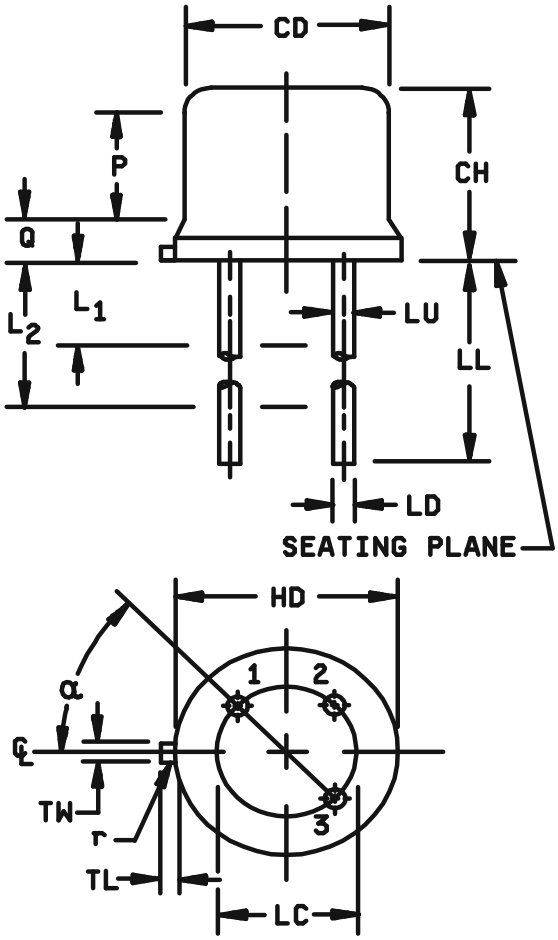
Symbol	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		
LD	.016	.021	0.41	0.53	4
LL	See notes 8 and 9				
LU	.016	.019	0.41	0.53	4
L_1	- - -	.050	- - -	1.27	
L_2	.250	- - -	6.35	- - -	
TL	.028	.048	0.71	1.22	5
TW	.036	.046	0.91	1.17	
P	.100	- - -	2.54	- - -	3, 6
Q	- - -	.040	- - -	1.02	
r	- - -	.007	- - -	0.8	
α	45°TP		45°TP		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between L_1 and L_2. LD applies between L_2 and 0.5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond 0.5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.
7. The collector shall be electrically connected to the case.
8. In accordance with ANSI Y14.5M, diameters are equivalent to Nx symbology.

FIGURE 1. Physical dimensions (TO - 18).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels only
11	h_{FE1}; I_{CBO1}
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CBO1} = 100 percent of initial value or 5 nA dc; whichever is greater; Δh_{FE1} = ±25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5).
$V_{CB} \geq 20$ V dc.
P_T = 300 mW dc.

NOTE: No heat sink or forced air cooling of the devices shall be permitted.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, appendix E, table V, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (endpoints) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E; table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750. V_{CB} = 20 V dc. P_T = 300 mW dc, no heat sink or forced air cooling of the devices shall be permitted.
B3	2037	Test condition A: all internal leads for each device shall be pulled separately.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E; table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection appendix E; table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E.
C6	1026	T_A = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5). V_{CB} = 20 V dc, P_T = 300 mW dc, no heat sink or forced air cooling of the devices shall be permitted.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E; table IX of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.4.1 Group E inspection appendix E; table IX of MIL-PRF-19500.

Subgroup	Method	Conditions
E1	1051	Condition C, 500 cycles; 22 devices, c = 0.
E1	1071	
E2	1038	80 to 85 percent of rated V_R, T_A = +150°C, t = 1,000 hours; 22 devices, c = 0.
E3		Not applicable
E4	3131	See 4.5.2, 22 devices, c = 0.
E5		Not applicable

4.5 Method of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750 (for qualification only). The following details shall apply:

a. Collector current magnitude during power applications shall be 28 mA dc.

b. Collector to emitter voltage magnitude ≥ 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 97°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage collector-to-base	3001	Bias condition D, I_C = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Breakdown voltage emitter-to-base	3026	Bias condition D, I_E = 10 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	6		V dc
Breakdown voltage collector-emitter	3011	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	45		V dc
Collector to emitter cutoff current	3041	Bias condition B, V_{CE} = 45 V dc	I_{CES1}		2.0	nA dc
Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 5 V dc	I_{CEO}		2.0	nA dc
Emitter-base cutoff current	3061	Bias condition D, V_{EB} = 5 V dc	I_{EBO}		5.0	nA dc
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 45 V dc	I_{CBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 µA dc, pulsed (see 4.5.1)	h_{FE1}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 500 µA dc, pulsed (see 4.5.1)	h_{FE2}	150		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 mA dc, pulsed (see 4.5.1)	h_{FE3}		600	
Collector-emitter saturated voltage	3071	I_C = 10 mA dc, I_B = 0.5 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Base-emitter saturated voltage	3066	Test condition A, I_C = 10 mA dc, I_B = 0.5 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)}$	0.6	1.0	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +175°C				
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 45 V dc	I_{CES2}		10	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 µA dc, pulsed (see 4.5.1)	h_{FE4}	20		
Subgroup 4						
Small signal short-circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc, f = 1 kHz, I_C = 1 mA dc:	h_{fe}	150	600	
Magnitude of common emitter small-signal short-circuit, forward-current transfer ratio	3306	V_{CE} = 5 V dc, I_C = 500 µA dc, f = 30 MHz	$\|h_{fe}\|$	1.5	6.0	
Open circuit output capacitance	3236	V_{CB} = 5 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8.0	pF
Noise figure	3246	V_{CE} = 5 V dc, I_C = 10 µA dc; R_g = 10 kΩ	NF			
Test 1		f = 100 Hz			5	dB
Test 2		f = 1 kHz			3	dB
Test 3		f = 10 kHz			3	dB
Small signal short-circuit output admittance	3216	V_{CB} = 5 V dc, I_E = 1.0 mA dc, f = 1 kHz	h_{ob}	0	1.0	µΩ
Small-signal open circuit reverse voltage transfer ratio	3211	V_{CB} = 5 V dc, I_E = 1.0 mA dc, f = 1 kHz	h_{rb}		$6x10^{-4}$	
Small-signal short-circuit input impedance	3201	V_{CB} = 5 V dc, I_E = 1.0 mA dc, f = 1 kHz	h_{ib}	25	32	Ω
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups B, C, and E electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 45 V dc	I_{CES}		2.0	nA dc
2.	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 45 V dc	I_{CES}		4.0	nA dc
3.	Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 µA dc, pulsed (see 4.5.1)	h_{FE1}	80	360	
4.	Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 10 µA dc, pulsed (see 4.5.1)	h_{FE1}	100	300	
5.	Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 10 mA dc, pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc

1/ The electrical measurements for appendix E; table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 4.

b. Subgroup 3, see table II herein, steps 2, 3, and 5.

c. Subgroup 6, see table II herein, steps 2, 3, and 5.

2/ The electrical measurements for appendix E; table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 4.

b. Subgroup 3, see table II herein, steps 1 and 4.

c. Subgroup 6, see table II herein, steps 2, 3, and 5.

3/ The electrical measurements for appendix E; table VII of MIL-PRF-19500 are as follows:

a. Subgroup 1, see table II herein, steps 1 and 4.

b. Subgroup 2, see table II herein, steps 1 and 4.

c. Subgroup 4, see table II herein, steps 1, 4, and 5.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI, SM
Navy - AS
Air Force - 13, 85, 99
NASA - NA

Preparing activities:
DLA - CC

(Project 5961-1691)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 5, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-PRF-19500/253E**	**2. DOCUMENT DATE (YYMMDD)** 970927

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER TYPE 2N930 JAN, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

5. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (514) 592-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) , Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43215-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3455 Telephone (703) 755-2340 AUTOVON 289-2340

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